SAT 9

Math Module 1

Question #	ID		Answer
SAT9 M 1.1	fd65f47f	Which expression is equivalent to $(2x^2 + x - 9) + (x^2 + 6x + 1)$? A. $2x^2 + 7x + 10$ B. $2x^2 + 6x - 8$ C. $3x^2 + 7x - 10$ D. $3x^2 + 7x - 8$	D
SAT9 M 1.2	6105234d	John paid a total of $\$165$ for a microscope by making a down payment of $\$37$ plus p monthly payments of $\$16$ each. Which of the following equations represents this situation? A. $16p - 37 = 165$ B. $37p - 16 = 165$ C. $16p + 37 = 165$ D. $37p + 16 = 165$	C
SAT9 M 1.3	ff2c1431	$7m = 5(n + p)$ The given equation relates the positive numbers m, n, and p. Which equation correctly gives n in terms of m and p? A. $n = \frac{5p}{7m}$ B. $n = \frac{7m}{5} - p$ C. $n = 5(7m) + p$ D. $n = 7m - 5 - p$	A
SAT9 M 1.4	a09ma104	The function g is defined by $g(x) = \sqrt{8x + 1}$. What is the value of $g(3)$? A $\frac{5}{8}$ B $\frac{25}{8}$ C 5 D 25	C

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Question #	ID		Answer
SAT9 M 1.5	47624288		C

The table gives the distribution of votes for a new school mascot and grade level for 80 students.

Mascot	Grade level			
	Sixth	Seventh	Eighth	Total
Badger	4	9	9	22
Lion	9	2	9	20
Longhorn	4	6	4	14
Tiger	6	9	9	24
Total	23	26	31	80

If one of these students is selected at random, what is the probability of selecting a student whose vote for new mascot was for a lion?

A. $\frac{1}{9}$

B. $\frac{1}{5}$

C. $\frac{1}{4}$

D. $\frac{2}{3}$

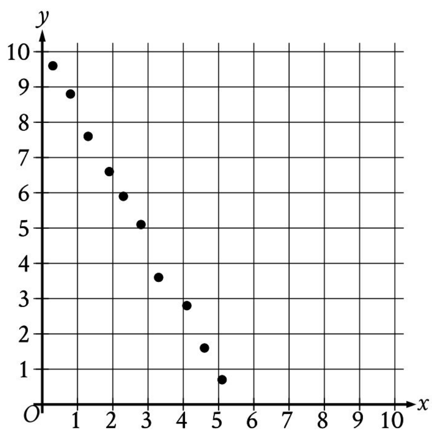
SAT9 M 1.6 a09ma106 **224**

$$f(x) = x + \frac{8}{11}$$

The function f is defined by the given equation.

What is the value of $f(x)$ when $x = \frac{3}{11}$?

SAT9 M 1.7 5f3ee607 **B**



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

A. $y = -1.9x - 10.1$

B. $y = -1.9x + 10.1$

C. $y = 1.9x - 10.1$

D. $y = 1.9x + 10.1$

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Question #	**ID**		**Answer**

SAT9 M 1.8 a09ma108 — **Answer: 1**

$$3x + 6 = 4y$$
$$3x + 4 = 2y$$

The solution to the given system of equations is (x, y). What is the value of y?

SAT9 M 1.9 a456cfd2 — **Answer: 14**

Data value	Frequency
6	3
7	3
8	8
9	8
10	9
11	11
12	9
13	0
14	6

The frequency table summarizes the 57 data values in a data set. What is the maximum data value in the data set?

SAT9 M 1.10 a09ma110 — **Answer: D**

Circle K has a radius of 4 millimeters (mm). Circle L has an area of 100π mm^2. What is the total area, in mm^2, of circles K and L?

A. 14π

B. 28π

C. 56π

D. 116π

Question #	ID		Answer
SAT9 M 1.11	a09ma111	If $9(4-3x)+2=8(4-3x)+18$, what is the value of $4-3x$? (A) -16 (B) -4 (C) 4 (D) 16	D
SAT9 M 1.12	bcb66188	Triangle FGH is similar to triangle JKL, where angle F corresponds to angle J and angles G and K are right angles. If $\sin(F)=\frac{308}{317}$, what is the value of $\sin(J)$? A. $\frac{75}{317}$ B. $\frac{308}{317}$ C. $\frac{317}{308}$ D. $\frac{317}{75}$	B
SAT9 M 1.13	9f6f96ff	A wire with a length of 106 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 6 more than 4 times the value of y. What is the value of x? A. 25 B. 28 C. 56 D. 86	D
SAT9 M 1.14	038d87d7	A neighborhood consists of a 2-hectare park and a 35-hectare residential area. The total number of trees in the neighborhood is $3{,}934$. The equation $2x+35y=3{,}934$ represents this situation. Which of the following is the best interpretation of *x* in this context? A. The average number of trees per hectare in the park B. The average number of trees per hectare in the residential area C. The total number of trees in the park D. The total number of trees in the residential area	A

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Question #	ID		Answer

SAT9 M 1.15 42f19012 — Answer: **B**

Which expression is equivalent to $a^{\frac{11}{12}}$, where $a > 0$?

A. $\sqrt[12]{a^{132}}$

B. $\sqrt[144]{a^{132}}$

C. $\sqrt[121]{a^{132}}$

D. $\sqrt[11]{a^{132}}$

SAT9 M 1.16 d135f4bf — Answer: **76**

The function f is defined by $f(x) = (x - 6)(x - 2)(x + 6)$. In the *xy*-plane, the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 4 units. What is the value of $g(0)$?

SAT9 M 1.17 1362ccde — Answer: **35**

$$y = 4x + 1$$
$$4y = 15x - 8$$

The solution to the given system of equations is (x, y). What is the value of $x - y$?

SAT9 M 1.18 1f353a9e — Answer: **D**

$$f(t) = 8{,}000(0.65)^t$$

The given function f models the number of coupons a company sent to their customers at the end of each year, where t represents the number of years since the end of 1998, and $0 \leq t \leq 5$. If $y = f(t)$ is graphed in the *ty*-plane, which of the following is the best interpretation of the *y*-intercept of the graph in this context?

A. The minimum estimated number of coupons the company sent to their customers during the 5 years was $1{,}428$.

B. The minimum estimated number of coupons the company sent to their customers during the 5 years was $8{,}000$.

C. The estimated number of coupons the company sent to their customers at the end of 1998 was $1{,}428$.

D. The estimated number of coupons the company sent to their customers at the end of 1998 was $8{,}000$.

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Question #	ID		Answer
SAT9 M 1.19	a09ma119	A landscaper uses a hose that puts $88x$ ounces of water in a bucket in $5y$ minutes. Which expression represents the number of ounces of water the hose puts in the bucket in $9y$ minutes at this rate?	D

(A) $\frac{9x}{440}$

(B) $\frac{440x}{9}$

(C) $\frac{5x}{792}$

(D) $\frac{792x}{5}$

SAT9 M 1.20 d0a53ef5 — Answer: **-3**

$$\sqrt{(x-2)^2} = \sqrt{3x+34}$$

What is the smallest solution to the given equation?

SAT9 M 1.21 3b225698 — Answer: **C**

Triangle XYZ is similar to triangle RST such that X, Y, and Z correspond to R, S, and T, respectively. The measure of $\angle Z$ is 20° and $2XY = RS$. What is the measure of $\angle T$?

A. 2°

B. 10°

C. 20°

D. 40°

SAT9 M 1.22 0e61101e — Answer: **B**

$$f(x) = 9(4)^x$$

The function f is defined by the given equation. If $g(x) = f(x+2)$, which of the following equations defines the function g?

A. $g(x) = 18(4)^x$

B. $g(x) = 144(4)^x$

C. $g(x) = 18(8)^x$

D. $g(x) = 81(16)^x$